

Protocol for creation of mold for microfluidics

1) Spin coat silane into glass slides for better adhesion:

- Done on the bench.
- Activate the vacuum pump and turn the slider for N2.
- We put slide of glass on it, activate vacuum, verify it sticks.
- Drop some drops of silane on it.
- Run program: **10seconds in 700rpm and 1min 4000 rpm.**
- **Change program: select program A and then modify with f1 and modify with add step and then the del step:-> select the f1 to restart from beginning**
- Let it rest at 100c for around 5min.

2) Prepare the platform:

- There are different frames that we can put above the glass. They have different thickness that will determine the thickness of the polymer layer we will coat on the top of the glass.
- Form the “sandwich”: glass -> frame -> photoresist LF-55GN.1 -> lamina of mylar sheet (use block to expand photoresist and have homogenous distribution).
- Place the chrome mask on top of it. Secure it and elevate the platform up until it is very tight.
- Can use the chrome mask there to make a dose test to tune duration if needed. For the 300-400 microns, around 250s-350s was initially tested to be overexposed and 100s slightly underexposed. So, expositions at around 150s could be envisioned for this height.
- Put under the uv light for the required time. (let the machine start and takes around 10min to do so).
- If it not enough time -> structures could collapse on polymer, and if too much type of pattern can look very curvy and undefined at edges.

3) Development:

- Recover: remove both upper and lower laminas. Go to bench and develop using soap and water. Develop until we no longer see white.
- Rinse with IPA and water and dry with nitrogen gun.
- Observe in microscope for sharpest edges and no collapses.

4) Second polymerisation and dehydration:

- Fill a petri dish with water and put our glass on it.
- Expose on UV for 10min at least (ideally more than 30min).
- Let it to dehydrate for around 2hours or more in a hotplate at around 70°C.